Spot the Misleading Chart

Student worksheet · Advanced level

Name:	Class:	Date:

Every chart tells a story. A good chart tells a **true** story you can see at a glance. Use this sheet with the *Good Charts vs Bad Charts* website, set to the same level.

Part 1: Match each mistake to its rule

Write the letter of the rule that fixes each mistake.

Mistake	Letter	Rule
1. Bar chart axis starting above zero		**A.** Space dates by how much time really passed.
2. Two y-axes stretched to match		**B.** Show the trend, not the noise.
3. Joining every weekly result with a line		**C.** Bar charts always start at zero.
4. Color suggesting a group that isn't there		**D.** Up on the chart should mean more.
5. Pie chart with nine slices		**E.** To compare more than a few things, use sorted bars.
6. Upside-down y-axis		**F.** Show each period, not the total so far.
7. Running total hiding a drop		**G.** A new color should mean a new group.
8. Uneven gaps between dates		**H.** Be suspicious when two lines match perfectly.

Part 2: Take the quiz

Do the quiz on the website. Write your score, then the rule you found hardest and why.

My score:	out of 11
Hardest rule and why:	

Part 3: When can the rules bend?

Explain in a sentence or two.

1. Why do bar charts need to start at zero, but line charts don't always?
2. When is comparing totals misleading? What should you compare instead?
3. A chart shows two lines moving together. Why doesn't that prove one causes the other?

Part 4: Investigate a real chart

Use a graph from *What's Going On in This Graph?* or the news.

Source (where it's from, and who made it):
Notice / wonder:
Honest headline:
Misleading headline someone could write from the same data:

Go deeper:

Question	What I noticed
Who made it, and why? What might they want you to believe?	
How much data is behind it? Small samples bounce around more.	
Total or rate? Would “per person” change the story?	
What's left out? Time range, groups, context?	
Cause or correlation? Does it actually show one thing causing another?	

Part 5: Redesign it

Sketch your improved chart with your honest headline as the title:

Explain your choices: chart type, axis, colors, title. Why does each one make the story clearer or more honest?